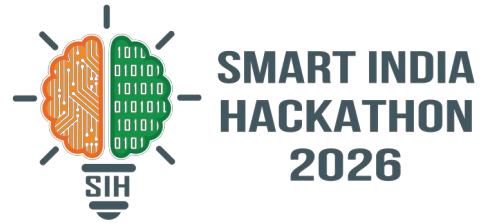


# SMART INDIA HACKATHON 2026



**Problem Statement ID –** SIH26183

**Problem Statement Title –** Real-Time Identification of Fraud-Linked Cryptocurrency Exchanges from Victim-Reported Suspect Wallet Addresses through Automated Blockchain Analytics

**Theme –** Blockchain & Cybersecurity

**PS Category –** Software

**Organisation –** Ministry of Home Affairs (I4C)

**Team ID –** 26183

**Team Name (Registered on portal) –** code2trip

**Institute –** code2trip Team



## PROPOSED SOLUTION

### ChainNetra

Autonomous Cyber-Forensic Intelligence Engine for Real-Time Exchange Identification, Asset Freezing & Courtroom Evidence Dossier Generation

Real-time blockchain forensics that traces stolen crypto across wallets, identifies the destination exchange, and auto-generates a court-ready freeze order — faster than any manual investigation.

#### PROPOSED SOLUTION: 4-STAGE CORE FORENSIC PIPELINE

##### 01 SUSPECT WALLET INTAKE

- Scans TRON, Ethereum & Bitcoin from a single suspect address
- Decodes hidden smart-contract token transfers (zero missed txns)
- SHA-256 timestamps every API response for tamper-proof custody

##### 02 MULTI-HOP FUND TRACING

- Autonomously follows funds through 10+ burner wallets across multiple hops
- Mathematically tracks exact stolen fraction across all splits
- Detects & removes wash-trade loops injected to mislead tracers

##### 03 EXCHANGE IDENTIFICATION

- Clusters deposit wallets to unmask the receiving exchange
- 7 pattern detectors flag peel chains, smurfing & rapid forwarding
- 4-tier confidence scoring: CONFIRMED / PROBABLE / POSSIBLE / UNATTRIBUTED

##### 04 LEGAL FREEZE & DOSSIER

- Maps exchange to its FIU-IND-designated Nodal LEO contact
- 1-click Section 94/106 BNSS freeze notice pre-filled with evidence
- Court-ready PDF with SHA-256 certified blockchain proof of funds

#### HOW IT ADDRESSES CRITICAL POLICING BOTTLENECKS

**The 'Golden Hour' Recovery Gap:** Fraudsters liquidate crypto via P2P within 1–2 hours. Manual tracing takes 6–12 hours. ChainNetra automates multi-hop graph traversal, enabling proactive freezing before fiat conversion.

**Overcoming Peeling & Smurfing Layering:** Syndicates split funds into micro-transfers across dozens of burner hops. ChainNetra's mathematical engine automatically tracks the exact fractional stolen value through every branch.

**Internal Smart Contract Visibility:** Scammers exploit DEX swaps and TRC-20 internal contract calls where public explorers display false zero balances. ChainNetra decodes execution traces natively with zero transaction drop.

**Eliminating Wrongful Freezes:** Strict 4-tier evidentiary confidence scoring ensures innocent merchants or payment gateways are never wrongfully frozen, protecting law enforcement from legal liability.

#### INNOVATION AND DOMAIN UNIQUENESS

**Deposit Wallet Reverse Attribution:** Detects high-frequency exchange deposit wallets without requiring internal exchange access, revealing the destination exchange directly from public blockchain patterns.

**Zero-Cost Architecture for Police Units:** Built 100% on open blockchain infrastructure and public intelligence, eliminating multi-lakh commercial software licenses for grassroots cyber cells.

**Automated Indian Legal Compliance:** Bridges technical blockchain evidence directly into Indian criminal procedure—auto-generating Section 94/106 BNSS freeze requisitions with pre-filled transaction hashes.

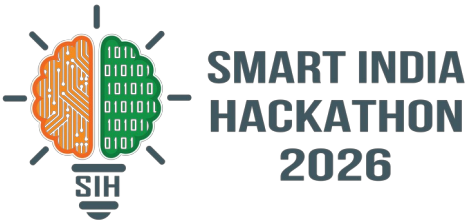
**Syndicate & Mule Network Correlation:** Cross-references separate victim complaints across states to discover shared deposit wallets, unmasking organized interstate cybercrime syndicates.

**Browser Addon for Field Investigators:** A Chromium extension lets officers check any wallet address on TronScan/Etherscan live — instantly overlaying ChainNetra's taint score without switching tools.

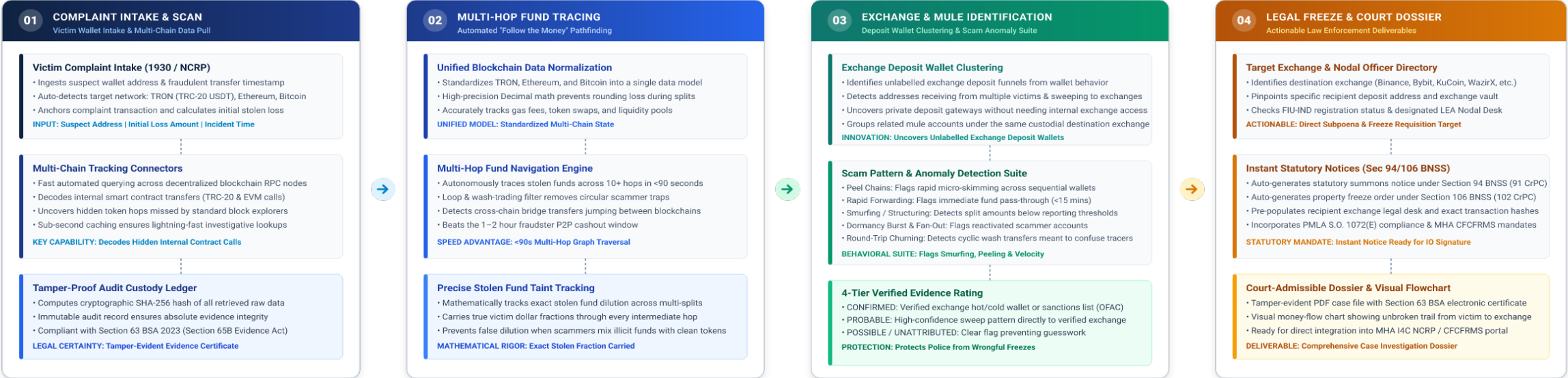
**Court-Admissible Cryptographic Custody:** Every external API response is SHA-256 hashed and stored with provenance, satisfying Section 63 BSA (Sec 65B IEA) electronic-evidence admissibility requirements.



# Technical Approach

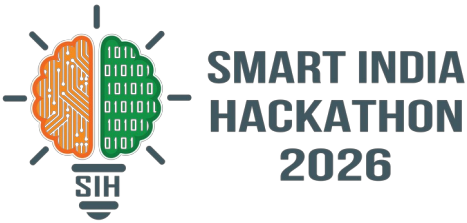


## METHODOLOGY & FORENSIC PIPELINE: AUTOMATED INVESTIGATION LIFECYCLE





# Feasibility & Viability



## OPERATIONAL, LEGAL & TECHNICAL FEASIBILITY OF CHAINNETRA

### TECHNICAL FEASIBILITY

#### Lightweight & Scalable

Runs on standard government cyber lab servers or state police data centers without requiring expensive specialized GPU/graph supercomputers.

### ECONOMIC FEASIBILITY

#### Zero Licensing Barrier

Leverages decentralized public RPCs, open-source intelligence (OSINT), and crowd-sourced verified label databases, requiring ₹0 software licensing fees.

### OPERATIONAL FEASIBILITY

#### Built for Grassroots IOs

Designed for investigating officers with zero blockchain engineering background—enter victim address, obtain actionable target VASP and pre-drafted freeze order.

### LEGAL FEASIBILITY

#### Courtroom Admissible

Every on-chain metric is mathematically deterministic, eliminating AI hallucinations and satisfying the strict electronic evidence requirements of Indian courts.

## REAL-WORLD CYBER-FORENSIC CHALLENGES

High-Velocity Peeling & Smurfing Layering: Fraudsters split illicit funds into dozens of sub-threshold transfers across 10+ hops to exhaust manual police tracing.

Internal Smart Contract & DEX Obfuscation: Over 90% of laundered USDT moves through contract calls and liquidity swaps that regular explorers fail to display.

Stale & Misattributed Public Address Labels: Inaccurate open-source labels risk pointing investigators toward the wrong entity or causing illegal freezes.

Commercial Payment Gateways Mimicking Scams: Legitimate e-commerce payment processors aggregate funds similarly to scam consolidation funnels.

Foreign Non-Compliant Exchanges (Offshore VASPs): Scammers deliberately funnel funds into non-FIU-IND compliant offshore exchanges that ignore Indian LEA notices.

## CHAINNETRA FORENSIC MITIGATION STRATEGIES

Autonomous Recursive Taint Engine: Traverses complex branching DAGs, carrying exact proportional stolen dollar value through every split regardless of depth or fan-out.

Native Smart Contract Trace Parsing: Decodes TRC-20 and EVM internal call trees directly, capturing hidden fund movements and bridge transfers.

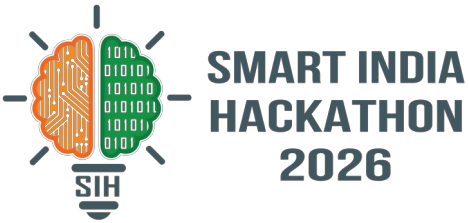
On-Chain Provenance & Verification: Every label is verified on-chain against historical volume and Proof-of-Reserves (PoR); unverified labels capped at PROBABLE/POSSIBLE.

Mandatory False-Positive Filtering: Behavioral suite actively checks for business hours, regular payroll sweeps, and merchant return loops before flagging.

FIU-IND Status Intelligence & Global LEA Routing: Flags VASP compliance status (FIU-IND registered vs. offshore), routing notices to international LEA portals (Interpol/NCB).



# Impact & Benefits



## SCALE OF CYBER FRAUD IN INDIA & OPERATIONAL VALUE DELIVERY

### ₹22,845+ Crore

**Annual Cyber Fraud Losses in India:** Official MHA data reported across Indian citizens on NCRP in 2024, highlighting the critical urgency for automated tracing tools.

### 22.68+ Lakh Cases

**Registered Cybercrime Incidents:** Complaints registered on the National Cybercrime Reporting Portal (+42% YoY surge), overwhelming district investigation units.

### Automated vs Manual

**Investigation Velocity:** Automated multi-hop graph traversal replaces days of manual blockchain explorer work — investigators get exchange identification instantly instead of 6–12 hours of manual analysis.

## TARGET USER AUDIENCE & OPERATIONAL VALUE DELIVERY

### DISTRICT IOs — Cyber Cell Sub-Inspectors

Empowers grassroots investigating officers with zero crypto training to trace suspect wallets and obtain ready-to-sign Section 94/106 BNSS (Sec 91/102 CrPC) notices within minutes.

### STATE CYBER CELLS — I4C & State Central Analysts

Discovers hidden syndicate clusters where the same exchange deposit funnel collects stolen funds from dozens of complaints across multiple states, unmasking organized crime networks.

### POLICE LEADERSHIP — Supervisory Chiefs & Courts

Delivers court-admissible Section 63 BSA (Sec 65B IEA) certified PDF case files with complete cryptographic SHA-256 hash trails for swift judicial conviction and asset recovery.

## OPERATIONAL WORKFLOW COMPARISON

### CURRENT MANUAL INVESTIGATION BOTTLENECK

Investigating officers manually toggle public block explorers across 20 browser tabs. Tracing 3 hops takes 6–12 hours, frequently stalls at internal contract calls or mixers, and yields unstandardized notes inadmissible in court.

### WITH CHAINNETRA AUTOMATION

Autonomous multi-hop tracing, DAR clustering, and VASP identification run fully automated. Instantly generates Section 94/106 BNSS (Sec 91/102 CrPC) legal freeze notices with SHA-256 evidence logs.

### ETHICAL, LEGAL & TECHNICAL BOUNDARIES OF CHAINNETRA

- Custodial Institution Identification, Not Private Doxxing: Public blockchain data identifies regulated custodial entities (VASPs); private citizen KYC remains protected until lawful subpoena execution.
- Objective Taint Evidence vs. Criminal Guilt: The system outputs mathematical contamination scores and behavioral anomaly metrics; formal guilt determination remains the sole prerogative of the court.
- Open Integration Standard: Designed with modular REST APIs ready for direct native integration into I4C NCRP and CFCFRMS portals.



## INDIAN STATUTORY FRAMEWORK, FORENSIC RESEARCH & COMPLIANCE ASSURANCE

### INDIAN STATUTORY & PROCEDURAL MANDATES

**MHA / I4C NCRP-CFCFRMS SOP (Jan 2026):** Establishes standard protocol directing LEAs to conduct forensic VDA tracing to identify custodial exchange nodes for non-onboarded VASPs.

**Section 94 BNSS 2023 (Sec 91 CrPC):** Statutory summons authority empowering investigating officers to requisition KYC records, login IP histories, and linked accounts from exchanges.

**Section 106 BNSS 2023 (Sec 102 CrPC):** Police power to issue immediate property seizure and balance freezing directives on fraud-linked VASP custodial accounts.

**Section 63 BSA 2023 (Sec 65B IEA):** Mandates cryptographic SHA-256 hash certification and immutable chain-of-custody for electronic evidence admissibility in Indian courts.

**PMLA Gazette S.O. 1072(E) & FIU-IND:** Mandates anti-money laundering registration and reporting compliance for all Virtual Digital Asset Service Providers operating in India.

### ACADEMIC FORENSICS & THREAT INTELLIGENCE

**UNODC Southeast Asia Crime Report (2024):** Identifies TRON-based USDT as the predominant currency for regional scam syndicates, confirming the necessity of specialized TRC-20 tracking.

**TRM Labs Illicit Crypto Report (2025):** Documents that 58%+ of global illicit stablecoin volume moves via TRON due to sub-dollar fees, validating ChainNetra's deep TRC-20 trace parsing.

**Enforcement Directorate (ED) Precedents:** Case studies (₹285 Cr scam, ₹391 Cr syndicate) prove illicit proceeds systematically off-ramp via P2P networks into Indian bank accounts.

**Victor (2020) Deposit Address Clustering:** Peer-reviewed heuristic establishing deposit-address-reuse patterns and sweeping thresholds for custodial exchange attribution.

**Meiklejohn et al. Taint Tracking Models:** Scientific validation proving Proportional Haircut Taint propagation is superior to FIFO in split, merge, and mixing laundering scenarios.

### SYSTEM INTEGRITY & PRODUCTION VERIFICATION

**Forensic Anomaly Suite Validation:** Tested against realistic synthetic laundering patterns: peel chains, rapid forwarding, smurfing, dormancy bursts, and wash churning.

**OFAC SDN Sanctioned Entity Ingestion:** Incorporates 400+ verified cryptocurrency addresses from global sanctions lists, automatically flagged as Tier-1 CONFIRMED sanctions risks.

**Proof-of-Reserves (PoR) Verification:** Exchange-published reserve addresses (Binance, OKX, CoinDCX, WazirX) ingested and verified on-chain to maintain zero-cost attribution precision.

**Cryptographic Chain-of-Custody:** Every raw JSON response from external RPCs is stored alongside its SHA-256 hash in PostgreSQL to ensure unbroken chain-of-custody in court.

**Designed for National Scale (NCRP / CFCFRMS):** Modular API architecture ready for direct integration into I4C's national cybercrime reporting infrastructure and state cyber cells.

Compliance Assurance: All blockchain queries adhere strictly to public API rate terms. No proprietary databases scraped. Every label records source, license, and capture timestamp.